Individual Contribution Report

Design and Creative Technologies

Torrens University, Australia

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1. Group Details

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- Source code: https://github.com/lfariabr/review-pulse
- Presentation: http://
- Video recording: http://
- Live Demo: https://review-pulse.streamlit.app/

2. Team Contribution Table

Member	Main contribution	%
Luis	Full technical implementation, app, tests, docs	65%
Samiran	Problem framing, live demo, future work	17.5%
Victor	Dataset, error analysis, ethics presentation	17.5%

3. Individual Report

My primary contribution to ReviewPulse was the full technical implementation of the ML pipeline – nine source modules, the Streamlit web app, 117 unit tests, and all project documentation.

On the data side, I built the pseudo-XML parser (*src/parser.py*), the preprocessing pipeline with negation expansion (*src/preprocess.py*), and the vocabulary builder and PyTorch DataLoaders (*src/dataset.py*). I then implemented both models: the TF-IDF + Logistic Regression baseline (*src/baseline.py*) and the bidirectional LSTM (Hochreiter & Schmidhuber,

1997) initialised with GloVe 100-dimensional embeddings (*src/model.py*). The training loop (*src/train.py*) uses Adam with gradient clipping, F1-based checkpointing, and Apple MPS device support. I implemented evaluation with confusion matrix and error analysis (*src/evaluate.py*), a unified inference API (*src/inference.py*), and the Streamlit web app with model selector and sample review generator (*app.py*). I also produced the presentation outline, 10 acceptance test cases with real model outputs, and this document.

An important ethical consideration is that the Blitzer et al. (2007) dataset uses filename-derived labels rather than human raters. Because the dataset uses filename-derived labels rather than direct human annotation, we audited for possible rating/text conflicts and ambiguous boundary cases. In this dataset, we found zero ambiguous rows, but the risk remains relevant in broader sentiment classification settings. BiLSTM confidence values are also uncalibrated – 98% confidence does not imply 98% accuracy – and the model generalises poorly to out-of-distribution text such as logistics reviews (Bender et al., 2021). Any production deployment requires human oversight and periodic label audits.

I estimate my contribution at 65% as the primary technical implementer. Victor contributed 17.5% covering dataset analysis, error analysis, and ethics. Samiran contributed 17.5% covering the problem framing, live demo delivery, and future work.

4. Appendices

4.1 Appendix A – Github Repository

Complete record of all model variants run across Tasks 1–4. All RMSE/MAE values are on the 41-row held-out test set.

Table 1. All model variants – Tasks 1–4 – with test-set RMSE and MAE. ***Bold = best result per task.***

Task	Model	Architecture	RMSE	MAE
1	Linear	-	$8,105	$4,729
1	MLP	[64]	$5,299	$3,108
1	MLP	[64, 32]	$5,338	$2,952
1	MLP	[128, 64]	$5,500	$3,078
1	Wide & Deep	[64, 32] + linear path	$5,873	$3,283
1	**RandomForest**	**500 trees**	**$2,873**	**$2,000**
2	MLP + Z-score	[64]	$14,109	$11,878
2	MLP + Min-Max	[64]	$13,299	$10,633
2	**RF + Z-score**	**500 trees**	**$2,875**	**$2,007**
3	Linear + One-Hot	-	$15,595	$13,215
3	MLP + One-Hot	[128, 64]	$2,812	$1,773
3	RF + One-Hot	500 trees	$3,913	$2,515
3	Linear + Ordinal	-	$15,290	$12,870
3	MLP + Ordinal	[128, 64]	$6,971	$4,178
3	RF + Ordinal	500 trees	$4,587	$2,900
4	Linear	-	$15,546	$13,208
4	MLP	[128, 64]	$3,235	$2,204
4	**RandomForest**	**500 trees**	**$2,777**	**$1,902**
4	Wide & Deep	[128, 64] + linear path	$4,223	$2,647

4.2 Appendix B – Live Application

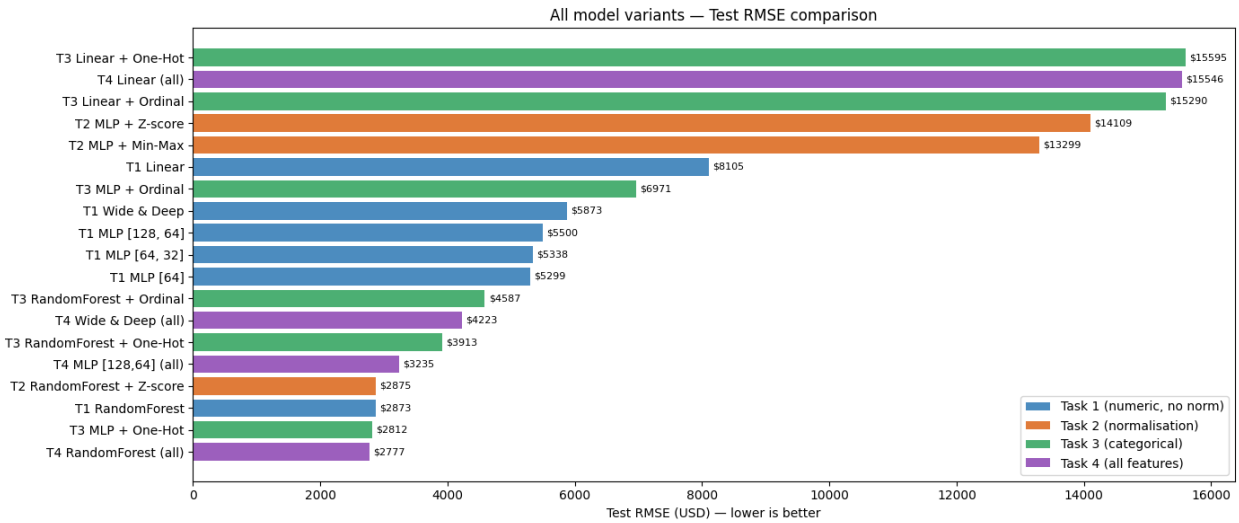

Figure 4. Horizontal bar chart of test-set RMSE for all model variants across Tasks 1–4, color-coded by task. Generated by the notebook. Lower bars = better.

3.3 Appendix C – Sample Test Cases Output

The raw UCI Autos CSV uses '?' as a placeholder for missing values. These were stored as *object dtype*, making numeric operations fail silently.

Steps applied:

1. **Convert '?' → NaN** via *pd.to_numeric(..., errors='coerce')* on all numeric columns.
2. **Drop rows where price is NaN or ≤ 0** – 4 rows removed, leaving 201 usable examples.
3. **Fill missing feature values** with the column mean – preserves rows rather than discarding them, which matters on a 205-row dataset.
4. **60/20/20 split** – *train_test_split(..., random_state=42)* → 120 train / 40 val / 41 test. All normalisation statistics computed on train only. Train label mean: $13,210. Test set mean: $13,088.

Train label mean: $13,210. Test set mean: $13,088.

3.4 Appendix D –Future Work

RF is the only architecture whose RMSE is deterministic across runs and stable across all four task configurations. This table directly supports its selection as the best overall model.

Table 2. RandomForest test-set RMSE across all four tasks – showing scale-invariance and feature-set robustness.

Task	Feature set	RMSE	MAE	Notes
T1	15 numeric, no norm	$2,873	$2,000	Baseline – no encoding, no scaling
T2	15 numeric + Z-score	$2,875	$2,007	Normalisation has zero effect on RF
T3	10 categorical, one-hot (55 cols)	$3,913	$2,515	Higher RMSE – numeric signal absent
T4	**70 features (15 numeric + 15 OHE)**	**$2,777**	**$1,902**	**Best overall – RF uses all features optimally**

The rapid early convergence (epoch 5→20 drops $2,300) reflects the *make* column acting as a near-lookup-table: one-hot encoding gives each brand a dedicated weight that the model adjusts independently.

End of Appendix Section

Statement of Acknowledgment

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- OpenAI ChatGPT (GPT–5.4)
- Anthropic Claude Sonnet 4.7

Both tools were used to assist with understanding ML concepts, structuring the technical manual, improving clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1. *"I built a BiLSTM sentiment classifier with pack_padded_sequence to stop the LSTM processing padding tokens. My training achieves val F1=84.0% at epoch 9, but test F1 drops to 80.3% while the TF-IDF baseline reaches 81.9% on the same held-out set. Can you help me understand why the neural model didn't generalise better despite the stronger validation score?"*
2. *"I need to compare two approaches for sentiment analysis – TF-IDF + Logistic Regression vs BiLSTM with GloVe – in an academic report where the simpler model wins on the held-out test set. How do I frame this honestly without undermining the neural architecture requirement of the assessment rubric?"*
3. *"Format this as APA 7th: Bender, Emily M., Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell, 2021, article titled On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?, conference ACM FAccT 2021, pages 610 to 623."*

I confirm that the use of these tools has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

5. References

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM FAccT Conference* (pp. 610–623). https://doi.org/10.1145/3442188.3445922

Blitzer, J., Dredze, M., & Pereira, F. (2007). Biographies, Bollywood, Boom-boxes and Blenders: Domain adaptation for sentiment classification. In *Proceedings of the 45th Annual Meeting of the ACL* (pp. 440–447). ACL. https://aclanthology.org/P07-1056/

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation, 9* (8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735